INTERNATIONAL CYBERSECURITY & DIGITAL FORENSICS ACADEMY (ICDFA)

SBT-DF204 Case Study 2: Reconstructing Chrome Web History

Case title: Reconstructing Suspected Drug Trafficking from Chrome Web History

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Course code	SBT-DF204 Computer Forensics Case Studies
Assessment	Case Study 2: Individual browser history investigation
Case title	Reconstructing Suspected Drug Trafficking From Chrome Web History
Evidence examined	Google Chrome `History` SQLite database, 196,608 bytes
Environment	Kali Linux virtual machine in VMware; `sqlite3` 3.46.1 (2024-08-13)
Report date	12/10/2026

1. Executive conclusion

I examined a Google Chrome History database and reconstructed 74 recorded visits that span 2022-04-19 13:54:18 UTC to 15:02:56 UTC. The records show a browser profile that searched for Craigslist, signed in, worked through Craigslist's posting workflow to a confirmation page (visits 11-19), and then opened the post's management page and the post itself, titled 'cheaper than Rx supplements - health and beauty - by owner' (visits 20-21). The same profile then opened a Gmail mailbox named unsub.fscs@gmail.com (visits 22-42) and two messages titled 'cheaper than Rx supplements - order' and 'cheaper than Rx supplements -- txid: 517b2156...afdc5' (visits 43 and 48). It followed an imgur link through a Google redirect page (visits 49-50) and saved `proof_of_payment.png` to `C:\Users\FSCS_User\Desktop\` (downloads record 1, 14:56:58 to 14:57:10 UTC). Between 14:58:59 and 15:02:05 UTC, the profile searched for a blockchain explorer, looked up the same transaction identifier (TXID) on Blockchain.com and mempool.space (visits 55-58 and 67-70), and used a Kraken account (visits 59-71).

These records fit the structure of the case hypothesis: an advertisement, email communication, a cryptocurrency payment reference and a saved image named as proof of payment. They do not show what the advertisement offered, what any message or image contained, whether a payment was made or received, or who operated the browser. I rate my confidence as high that this browser profile performed the recorded navigation sequence, moderate that the sequence relates to an advertisement-and-payment workflow, and low for identifying any person or for confirming that any transaction was completed.

3. Evidence acquisition and integrity

I acquired the supplied Chrome `History` file on my Kali Linux virtual machine and preserved it before I examined any record (Figures 1-3).

Attribute	Recorded value
Source	`https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Digital_Currency/LabFiles/History`
Original file name	`History`
Acquired	2026-10-09, 10:44:43 to 10:44:48 (VM clock), by `wget`, HTTP 200
Size	196,608 bytes (192K) in `wget` and `ls -lh`
File type	`file`: SQLite 3.x database, last written using SQLite version 3036000, file counter 33, database pages 41, cookie 0xa7, schema 4, UTF-8, version-valid-for 33
SHA-256	`b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19`
Read-only original	`~/webhistory/History` (mode 444)
Working copy	`~/webhistory/History_working`
Tools	`wget`, `ls`, `file`, `sha256sum`, `chmod`, `cp`, `sqlite3` 3.46.1

I created the folder ~/webhistory, downloaded the file with wget and confirmed the transfer. Before I opened the database, I ran ls -lh to record the size, file to confirm the SQLite 3.x format and sha256sum to record the hash. I then ran chmod 444 History to make the original read-only, copied it with cp History History_working, and ran sha256sum on both files. The two values were identical, so I knew the copy was exact. I opened only History_working in sqlite3, which kept the original free of any database connection and any journal or metadata writes. I did not visit any URL, account, message, wallet address or transaction ID that appears in the history; I worked only from the database records.

Derived output. I exported the full visit timeline to timeline_full.csv (75 lines: one header and 74 visits) and hashed it:
54f50652d400580d519dbdd3c7d61784c246019d86f34709d10106b52ffea4c2 (Figure 16).

4. Database method

4.1 Tables reviewed

I listed the catalogue with SELECT name FROM sqlite_master WHERE type = 'table' ORDER BY name; (Figure 4). It returned 17 tables: clusters, clusters_and_visits, content_annotations, context_annotations, downloads, downloads_reroute_info, downloads_slices, downloads_url_chains, keyword_search_terms, meta, segment_usage, segments, sqlite_sequence, typed_url_sync_metadata, urls, visit_source and visits. I examined five of them for this case: urls, visits, downloads, downloads_url_chains and keyword_search_terms.

4.2 Fields examined

I ran PRAGMA table_info() on urls, visits and downloads to read each structure before I queried it (Figures 5, 7 and 12).

Table	Fields	Role in this case
urls	id, url, title, visit_count, typed_count, last_visit_time, hidden (I used id, url, title)	One row per distinct address; supplies the URL and page title
visits	id, url, visit_time, from_visit, transition, segment_id, visit_duration, incremented_omnibox_typed_score, opener_visit (I used id, url, visit_time, from_visit, transition, visit_duration)	One row per visit; supplies time, referrer visit, navigation type and dwell time
downloads	27 fields from id to embedder_download_data (I used id, target_path, total_bytes, state, danger_type, interrupt_reason, start_time, end_time, tab_url, mime_type, hash)	One row per saved file
downloads_url_chains	id, chain_index, url	The address chain a download came from
keyword_search_terms	url_id, term	Search text tied to a urls row

4.3 Timestamp conversion

Chrome stores times as WebKit timestamps, which count microseconds since 1 January 1601 UTC. I converted every timestamp with this expression (Figure 8):

```
datetime(visit_time/1000000 + strftime('%s','1601-01-01'), 'unixepoch')
```

`strftime('%s','1601-01-01')` returns -11,644,473,600, the number of seconds between 1601 and the Unix epoch, so the sum is a Unix time that `datetime` formats as UTC. I validated the method on visit 1, which returned 2022-04-19 13:54:18 UTC and matches 09:54:18 EDT in the course materials. I treated `visit_duration` as microseconds and divided by 1,000,000.0 to report seconds. The course slide for the visits table labels these fields as nanoseconds, but the lab brief and my validation both confirm microseconds. I report UTC in the main timeline. Where I give local time, I state it separately as US Eastern Daylight Time (UTC-4), the zone the course materials use; the History file does not record the device's time zone.

4.4 Joins

I joined `visits.url` to `urls.id` to attach each visit to its address and title. `visits.from_visit` points to another `visits.id` and records the referring visit. I joined `keyword_search_terms.url_id` to `urls.id` to read search terms. I read the downloads tables separately and tied them to browsing by the `tab_url` field and the time window.

4.5 Transition decoding

I decoded each `transition` value in two parts (Figure 9). First, I isolated the base type with `transition & 255`. Only three base types occur in this database: 0 (link) on 63 visits, 5 (omnibox-generated search) on 4 visits (1, 22, 51 and 59), and 7 (form submit) on 7 visits (8 and 14-19). Second, I read the qualifier bits in the upper part of the value:

Bit	Meaning	Where I found it
0x02000000	Typed from the address bar	Visits 22, 51, 59 (`0x32000005`)
0x10000000	Navigation chain start	Visits 3, 6, 11, 24
0x20000000	Navigation chain end	Visits 5, 7 (with 0x80000000) and 42, 46 (with 0x40000000); with the start bit (`0x30000000`) on single-visit chains such as 2 and 20
0x40000000	Client redirect	Visits 44, 45, 68
0x80000000	Server redirect	Visits 4, 12, 25

Visit 1 (`0x30000005`) lacks the address-bar bit that visits 22, 51 and 59 carry; I report the values as found and draw no conclusion from the difference. My first export printed some values as `0xFFFFFFFF80000000` because SQLite stores the field as a signed number and `printf` extended the sign. I masked the value with `& 4294967295` (0xFFFFFFFF), and the output became `0x80000000` and `0xA0000000`, which decode as server redirect and server redirect plus chain end.

4.6 Segmenting the visits

I split the 74 visits into navigation chains by counting the chain-start bit with `SUM((transition & 268435456) != 0) OVER (ORDER BY visit_id)`, which produced 44 chain numbers (Figure 10). I preferred this to the course method of cutting wherever `visit_duration = 0`, because zero duration also occurs on non-redirect visits such as 21, 22 and 47. The method has one weakness: it assumes chains do not interleave, and in Gmail they do. Chain 44 groups visit 71 (Kraken ledger, referrer visit 63) with visits 72-74, which descend from visit 48 through `from_visit` (74 to 73 to 72 to 48). I resolved this by following `from_visit` for every disputed visit.

5. Findings

Question 1: What evidence file did I acquire and how did I preserve it?

I acquired `History` (196,608 bytes) from the raw.githubusercontent.com source in Section 3, recorded its SHA-256 value, made the original read-only and analysed an identical working copy whose hash I verified (Figures 1-3). Evidence log: E01.

Question 2: Which tables and fields are relevant?

The `urls` table gives me the address and title of each distinct page, and the `visits` table gives me each visit with its time, referrer (`from_visit`), transition and duration (E03). `keyword_search_terms` ties four visits to search terms (E04). `downloads` and `downloads_url_chains` record the saved file and the address it came from (E09). Section 4.2 lists the fields I used.

Question 3: When did the recorded activity occur?

All 74 visits fall on 2022-04-19 between 13:54:18 and 15:02:56 UTC (09:54:18 to 11:02:56 EDT, assuming UTC-4). The record shows two gaps: nothing is logged between 13:59:31 and 14:04:12 (4 minutes 41 seconds) or between 14:05:59 and 14:56:47 (50 minutes 48 seconds). The first block (13:54:18 to 14:05:59) covers the Craigslist posting and the Gmail sign-in and inbox; the second (14:56:47 to 15:02:56) covers the payment-related activity. I used the conversion in Section 4.3 and cross-checked visit 1 against the course materials (E03; Appendix B).

Question 4: How did the browser reach the material URLs?

- **Address-bar searches.** Visits 1, 22, 51 and 59 have base type 5, and `keyword_search_terms` confirms the terms 'craigslist', 'gmail', 'blockchain explorer' and 'kraken' on url IDs 1, 17, 40 and 45 (E04; Figure 11).
- **Form submits.** Visit 8 is the Craigslist login form, and visits 14-19 are the six posting steps (base type 7).
- **Redirect chains.** I identified server-redirect chains at visits 3-5, 6-7, 9-10, 11-13, 24-30, 31-33, 37-42, 49-50, 55-56 and 67-70, and client-redirect activity at visits 43-46 and 72-74.
- **Unbroken path.** From visit 3 to visit 20, each `from_visit` names the previous visit, which gives one continuous path from the Craigslist home page to the posting management page. Visit 21 (the post itself) has `from_visit` 0, so the history records no referrer for it.

- **Explorer pages.** Visits 57 and 58 both have `from_visit` 54. I therefore state that the address page (visit 58) was reached from the explorer page, not from the transaction page.

Question 5: What evidence relates to the suspected sale or posting?

Visit	UTC	URL ID	Page	Transition	Duration (s)
8	13:54:39	5	accounts.craigslist.org/login/home, title 'craigslist account'	0x30000007	23.44
11	13:55:05	7	post.craigslist.org/c/bal, 'choose type'	0x10000000	0.00
12	13:55:05	8	post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5, 'posting confirmation'	0x80000000	0.00
13	13:55:05	9	same path, `?s=type`, 'choose type'	0xA0000000	4.27
14	13:55:10	10	`?s=cat`, 'choose category'	0x30000007	5.60
15	13:55:15	11	`?s=edit`, 'create posting'	0x30000007	177.67
16	13:58:13	12	`?s=geoverify`, 'add map'	0x30000007	4.39
17	13:58:17	13	`?s=editimage`, 'create posting'	0x30000007	1.65
18	13:58:19	14	`?s=preview`, 'create posting'	0x30000007	3.20
19	13:58:22	8	`/bQTi5`, 'posting confirmation'	0x30000007	4.99
20	13:58:27	15	post.craigslist.org/manage/7473121658, 'manage posting'	0x30000000	5.60
21	13:58:31	16	baltimore.craigslist.org/hab/d/baltimore-cheaper-than-rx-supplements/7473121658.html	0x30000000	0.00

I read these records as a posting workflow: type selection, category, content editing, location, image step, preview and confirmation (Figure 6 shows the matching `urls` rows). The same posting key (`sHbOUOi_7BG68qUDIbeTbQ/bQTi5`) runs through urls 8-14, each step names the previous visit in `from_visit`, and visit 15 lasted 177.67 seconds (13:55:15 to 13:58:13), which is consistent with entering text. The management page (url 15) and the ad (url 16) both carry post number 7473121658, and the ad's title reads 'cheaper than Rx supplements - health and beauty - by owner -...'.

Two points limit this evidence. First, the title 'posting confirmation' also appears at visit 12, during a redirect at 13:55:05 before any content step, so I relied on the sequence (preview at visit 18, confirmation at 19, management page at 20, ad at 21) and not on the title alone. Second, History stores addresses and titles, not what was typed into the form, so I cannot show the ad's description, price, location or images, or who could see it. Evidence log: E05 and E06.

Question 6: What evidence relates to communications and payment?

- **Gmail access (visits 22-42, E07).** After the omnibox search for 'gmail' (visit 22), redirect chains led through Google's sign-in pages: the identifier page (visit 35, 13:58:44), the password-challenge page (visit 36, 13:59:26) and the cookie and session pages `CheckCookie`, `SetOSID` and `SetSID` (visits 37-39). The titles name the mailbox unsub.fscs@gmail.com. Inbox titles

already appear at visits 25-26 (13:58:40), before the sign-in pages, so the titles alone do not tell me whether an existing session or the later sign-in gave access.

- **Messages (visits 43-48, E08 and E09).** At 14:04:12 the profile opened the message 'cheaper than Rx supplements - order' (url 34), then two `?compose=` URLs (urls 35 and 36, visits 44 and 45), then returned to the message and the inbox. At 14:56:47 it opened the message 'cheaper than Rx supplements -- txid: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5' (url 37; Figure 14).
- **Imgur and download (visits 49-50, downloads 1, E09).** Visit 49 is a `google.com/url?q=https://imgur.com/dTgrkP7` redirect page at 14:56:52, five seconds after the txid message. Visit 50 is the imgur page (url 39, 18.93 seconds). Download 1 started at 14:56:58 and ended at 14:57:10 UTC, inside that visit: target `C:\Users\FSCS_User\Desktop\proof_of_payment.png`, 33,844 bytes, `image/png`, state 1, danger type 0, interrupt reason 0, `tab_url` `https://imgur.com/dTgrkP7`, source chain `https://i.imgur.com/dTgrkP7.png` (Figure 13). The `hash` field is empty.
- **Blockchain explorer (visits 51-58, E10).** The profile searched for 'blockchain explorer', opened blockchain.com/explorer (18.68 seconds), searched for the TXID (visit 55), opened the transaction page (url 43) and then the address page for 38RcsURWCDmCbYocmsZCnGz1FtD7x477mt (url 44, 14:59:21; Figure 11).
- **Kraken (visits 59-71, E11).** After the omnibox search for 'kraken', the profile opened the sign-in page and then the device-approval page (visit 63, 104.43 seconds), `/u/trade`, `/u/instant` and `/u/funding` (url 50, title '41,543.80 USD - Kraken - Funding - Overview'). At 15:01:31 a Kraken redirect chain (visits 67-70) led to `mempool.space/tx/517b...`, and at 15:02:05 the profile opened the ledger page (url 55, title '41,563.70 USD - Kraken - History - Ledger'; Figure 15). I record these titles as observed and draw no link between the two figures and any payment.

The same 64-character TXID `517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5` appears in five page titles across three services (E13; Figures 14 and 15):

URL ID	Visit	UTC	Page
37	48, 74	14:56:47; 15:02:56	Gmail message 'cheaper than Rx supplements -- txid: ...'
42	55	14:59:06	Blockchain Explorer, 'Transaction: ...' (from a search URL)
43	56, 57	14:59:06	Blockchain Explorer, `btc/tx/` page
51	67	15:01:31	Kraken redirect, 'Transaction: ... - mempool - Bitcoin Explorer'
54	70	15:01:33	mempool.space `tx/` page

This repetition ties the email subject to the later explorer and Kraken activity. It does not show that the transaction exists, what it paid, or who owns the address.

Questions 7 and 8

I answer Question 7 in Section 6 (timeline) and Question 8 in Section 7 (attribution assessment).

6. Timeline

Times are UTC on 2022-04-19. Appendix B lists all 74 visits.

Stage	UTC	Visit or URL ID	Artifact	Observed fact	Interpretation and limit
1. Search for Craigslist	13:54:18-13:54:19	Visits 1-2; url 1	Google results page; `keyword_search_terms`	Omnibox search (`0x30000005`), then the results page stayed open 0.727 s; term 'craigslist'	I infer a search from the address bar. It does not show who typed it.
2. Reach Craigslist Baltimore	13:54:19	Visits 3-5; urls 2-4	craigslist.org, geo.craigslist.org, baltimore.craigslist.org	Chain `0x10000000`, `0x80000000`, `0xA0000000`; 2.47 s on the final page	The server redirected to the Baltimore site. A regional redirect does not show the device's physical location.
3. Sign in to Craigslist	13:54:22-13:54:39	Visits 6-8; urls 5-6	accounts.craigslist.org login pages	Login redirect (visit 7, 17.46 s on url 6), then a form submit (`0x30000007`, visit 8, 23.44 s) that landed on 'craigslist account'	Consistent with a completed sign-in. Credentials and the outcome are not stored.
4. Open the posting wizard	13:55:03-13:55:05	Visits 9-13; urls 2, 4, 7-9	post.craigslist.org/c/bal	Home page redirect (visits 9-10), 'choose type' page, redirect through the confirmation URL, then `?s=type` (4.27 s)	Marks the start of a new posting. The pass through the confirmation URL is a redirect artefact, not proof of publication.
5. Complete the posting form	13:55:10-13:58:22	Visits 14-19; urls 10-14, 8	`?s=cat`, `?s=edit`, `?s=geoverify`, `?s=editimage`, `?s=preview`, confirmation	Six form submits (`0x30000007`) in order; visit 15 lasted 177.67 s	Matches the Craigslist posting steps ending in a confirmation. The dwell time suggests content entry, but the entries are not recorded.
6. Manage and view the post	13:58:27-13:58:31	Visits 20-21; urls 15-16	`manage/7473121658`; ad page	Both URLs carry post number 7473121658; ad	The profile opened the management page and the ad. The

				title 'cheaper than Rx supplements - health and beauty - by owner -...'; visit 21 has `from_visit` 0	ad text, price and visibility are unknown.
7. Access Gmail	13:58:38-13:59:31	Visits 22-42; urls 17-33	Google search 'gmail'; sign-in and inbox pages	Omnibox search (`0x32000005`); chains 24-30, 31-33, 37-42; identifier page (visit 35) and password-challenge page (visit 36); titles name unsub.fscs@gmail.com	The profile reached that mailbox. Inbox titles at visits 25-26 precede the sign-in pages, so I cannot say which session gave access.
8. Open a message and compose windows	14:04:12-14:05:59	Visits 43-47; urls 33-36	Message 'cheaper than Rx supplements - order'	Message opened after a 4 min 41 s gap; two `?compose=` URLs (visits 44-45, `0x40000000`); return to the message (46) and inbox (47)	The profile opened compose windows on that message. I do not know whether any reply was written or sent.
9. Second message, imgur, download	14:56:47-14:57:10	Visits 48-50; urls 37-39; download 1	txid message; imgur page; `proof_of_payment.png`	Message opened after a 50 min 48 s gap; imgur page at 14:56:52 (18.93 s); download 14:56:58-14:57:10, 33,844 bytes, from `i.imgur.com/dTgrkP7.png`; hash empty	The link plausibly came from the message (Google redirect page five seconds later). I cannot see the image, so I cannot confirm it is a payment receipt.
10. Blockchain lookups	14:58:59-14:59:21	Visits 51-58; urls 40-44	Google search; blockchain.com pages	Omnibox 'blockchain explorer' (`0x32000005`); explorer page 18.68 s; TXID search, then `btc/tx` page; address page at 14:59:21 with `from_visit` 54	The profile looked up the identifier that also appears in the message subject. I did not query the live chain, so existence, amount and ownership are unverified.
11. Kraken	14:59:38-	Visits 59-	Google search	Sign-in page;	The profile used a

session	15:02:05	71; urls 45-55	'kraken'; kraken.com pages	device-approval page (visit 63, 104.43 s); `/u/trade`, `/u/instant`, `/u/funding` (title 41,543.80 USD); redirect chain 67-70 to `mempool.space/tx/517b...`; ledger page (visit 71, title 41,563.70 USD)	Kraken account and opened the same transaction through Kraken's redirect link. The titles show balance-like figures that I cannot tie to any payment.
12. Return to Gmail	15:02:32-15:02:56	Visits 72-74; urls 56, 57, 37	Compose URLs; txid message	Two more `?compose=` URLs (`0x40000000`), then the txid message (`0x60000000`); `from_visit` leads back to visit 48	The last recorded activity returns to the txid message. Compose URLs do not prove a sent message. Visits 72-73 carry titles that begin like the 'order' message although they descend from visit 48; I could not explain this.

7. Attribution assessment

7.1 Confidence

Statement	Confidence	Basis
The History file I analysed is intact and I examined it without altering it	High	Identical SHA-256 for the original and the working copy; original set read-only (E01)
This browser profile performed the recorded navigation sequence	High	74 visits linked by `from_visit` and chain flags, and a downloads record whose time window sits inside visit 50 (E03, E09)
The sequence relates to an advertisement-and-payment workflow	Moderate	A posting workflow, an email subject carrying a TXID, and the same TXID in explorer and Kraken titles (E05, E13); no page content
A named person operated the profile	Low	Only a Windows account name (`FSCS_User`) in the download path and a

Statement	Confidence	Basis
		mailbox name in page titles
A payment was made or received	Low	No amounts, no image content and no live-chain check

7.2 Limitations

1. **Profile versus person.** History records a Chrome profile. The Windows account name `FSCS_User` appears only in the download path, and the mailbox name unsub.fscs@gmail.com appears only in page titles. Neither identifies a person, and no credentials are recorded.
2. **Titles and URLs are not content.** I cannot see the advertisement's text, the messages, or the imgur image. The `hash` field for download 1 is empty, so I cannot verify the saved file, and the file itself is not part of my evidence.
3. **Compose URLs do not prove a sent message.** Visits 44, 45, 72 and 73 show that compose windows were opened. They do not show that anything was written or sent.
4. **The transaction and the wallet are unverified.** I did not query the live blockchain, so I cannot confirm that the TXID exists, what it paid, or who owns address 38RcsURWCDmCbYocmsZCnGz1FtD7x477mt. The course slides describe the address as the seller's, but the History file holds only the ad's title, not its body, so that statement is outside what my evidence supports. The address page (visit 58) has `from_visit` 54, which is the explorer page.
5. **Kraken titles.** Url 50 reads '41,543.80 USD - Kraken - Funding - Overview' and url 55 reads '41,563.70 USD - Kraken - History - Ledger', a difference of 19.90 USD. I report both as titles and do not tie them to any payment.
6. **History records page loads, not every action.** Clicks, typing and scrolling inside a page are not recorded, and single-page applications such as Gmail (the `#inbox` fragments) can hide in-page activity. Cached loads may also leave no separate visit.
7. **Gaps and completeness.** I found no visits between 13:59:31 and 14:04:12 or between 14:05:59 and 14:56:47. Chrome history can be cleared, and private-window activity is not stored here, so I cannot tell whether these are idle periods or missing records.
8. **Time zone.** I assumed US Eastern Daylight Time (UTC-4) for local times because the course materials use it. The file does not record the device's time zone.
9. **Method limits.** My chain numbering assumes chains do not interleave, and I corrected it with `from_visit` where Gmail broke that assumption (Section 4.6). I could not explain why visits 72-73 carry titles that begin like the 'order' message although they descend from visit 48.

7.3 Alternative explanations

- **A different person used the same profile or Windows account.** Shared access, remote access or a second user would produce identical records.
- **The profile owner was checking an incoming payment, not making one.** The same sequence fits either role, and the history does not separate them.

- **The activity was unrelated to a sale.** The same pages could appear if someone were researching these services. The repeated TXID and the post-and-email sequence make this less likely, but I cannot exclude it from History alone.

7.4 Defensible conclusion

On the evidence I examined, I conclude that the Chrome profile in this History file posted to and viewed a Craigslist ad at about 13:58 UTC on 2022-04-19, accessed the mailbox unsub.fscs@gmail.com, opened two messages about that ad (one with a TXID in its subject), saved an image named `proof_of_payment.png` from imgur, and then looked up the same TXID on Blockchain.com and mempool.space while using a Kraken account. I cannot conclude from History alone what was sold, that a payment occurred, who owned the address, or who operated the browser. To close those gaps I would examine the saved image and its hash, the Gmail message content, the Craigslist post content, Kraken account records, and Windows profile and sign-in artefacts for the `FSCS_User` account.

8. Appendices

Appendix A: Screenshot evidence

All screenshots are from my Kali Linux terminal. Figure 10 shows the query and its first rows; the complete output is in `timeline_full.csv` and reproduced in Appendix B.

Figure	Content	Evidence
1	`wget` download of `History`	E01
2	`ls -lh`, `file` and `sha256sum` on the original	E01
3	`chmod 444`, `cp` and the matching hashes for original and working copy	E01
4	Table list from `sqlite_master`	E02
5	`PRAGMA table_info(urls)`	E02
6	`urls` listing (ids 1-16)	E02, E05
7	`PRAGMA table_info(visits)`	E02
8	First `visits` and `urls` join with timestamp conversion	E03
9	Transition decoding query and its first row	E03
10	Chain-numbered timeline query and its first rows	E03
11	`keyword_search_terms` result and full URLs for urls 16, 39, 43, 44	E04, E06, E09, E10
12	`PRAGMA table_info(downloads)`	E02
13	`downloads` record and `downloads_url_chains`	E09
14	`downloads` record and full titles for urls 16, 34, 37	E09, E13
15	Full titles for urls 42-55 and the CSV export command	E13, E14
16	`timeline_full.csv` line count and SHA-256	E14

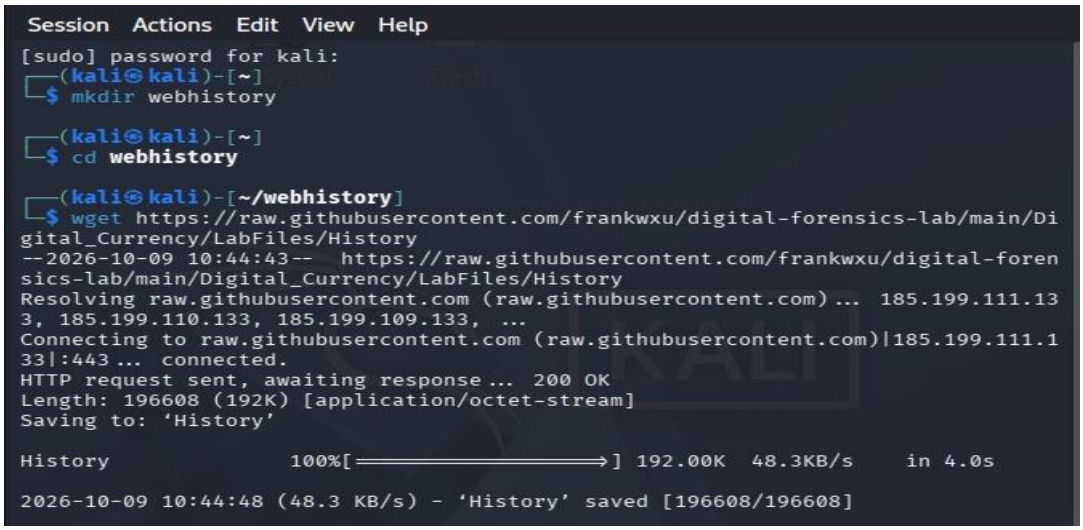


Figure 1. Acquiring the History file with wget in ~/webhistory.

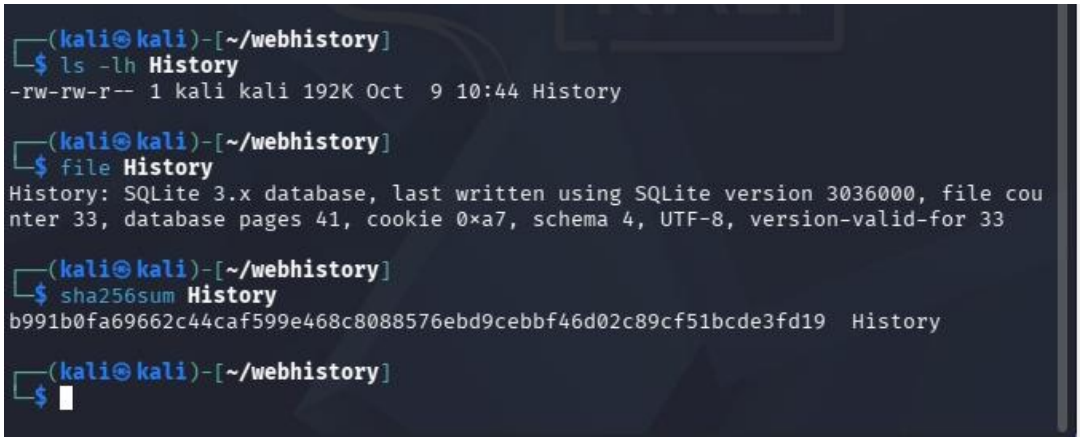


Figure 2. File size, file type and SHA-256 of the original History file.



Figure 3. Original set read-only, working copy created, and both hashes identical.

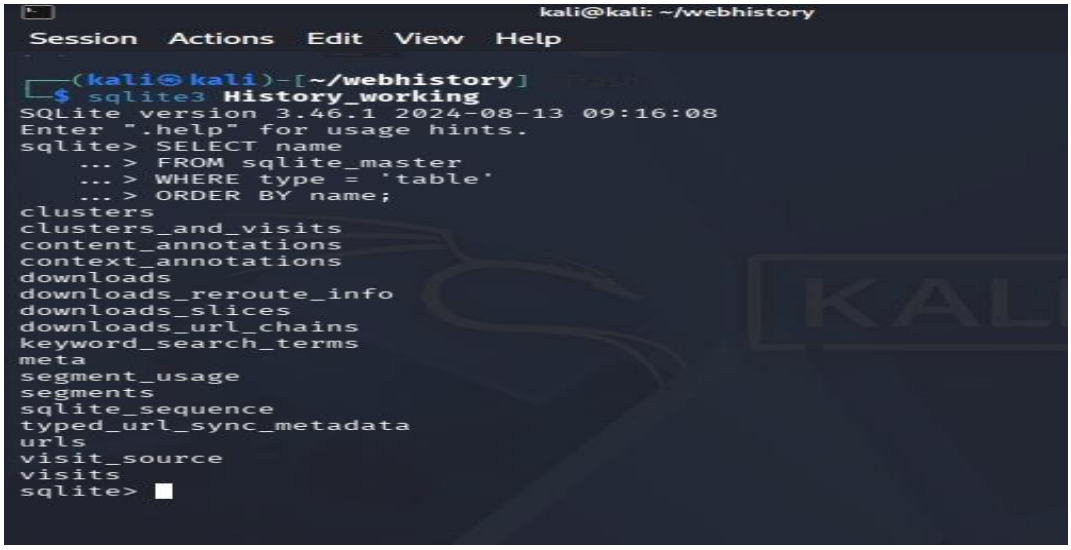


Figure 4. The 17 tables in History_working, listed from sqlite_master.



Figure 5. Structure of the urls table.

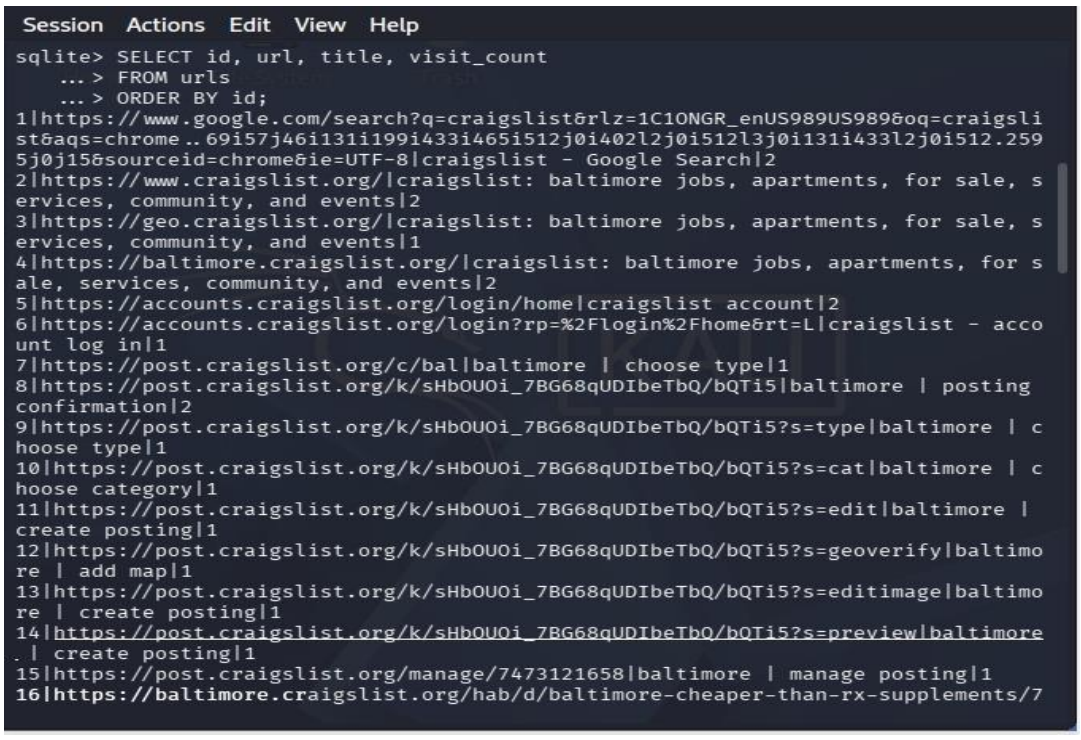


Figure 6. The urls table, ids 1-16, showing the Craigslist posting steps.

```
sqlite> PRAGMA table_info(visits);
0|id|INTEGER|0||1
1|url|INTEGER|1||0
2|visit_time|INTEGER|1||0
3|from_visit|INTEGER|0||0
4|transition|INTEGER|1|0|0
5|segment_id|INTEGER|0||0
6|visit_duration|INTEGER|1|0|0
7|incremented_omnibox_typed_score|BOOLEAN|1|FALSE|0
8|opener_visit|INTEGER|0||0
sqlite>
```

Figure 7. Structure of the visits table.

```
Session  Actions  Edit  View  Help
sqlite> .headers on
sqlite> .mode column
sqlite> SELECT
    v.id AS visit_id,
    u.id AS url_id,
    datetime(v.visit_time / 1000000 + strftime('%s', '1601-01-01'), 'unixepoch') AS
visit_utc,
    v.from_visit,
    v.transition,
    v.visit_duration / 1000.0 AS duration_ms,
    u.title,
    u.url
FROM visits AS v
JOIN urls AS u ON v.url = u.id
ORDER BY v.visit_time ASC;
visit_id  url_id  visit_utc           from_visit  transition   duration_ms  title
                                                    url

________  ______  ___________________  __________  __________  ___________  _______
_____________________________________________  ________________________________
_____________________________

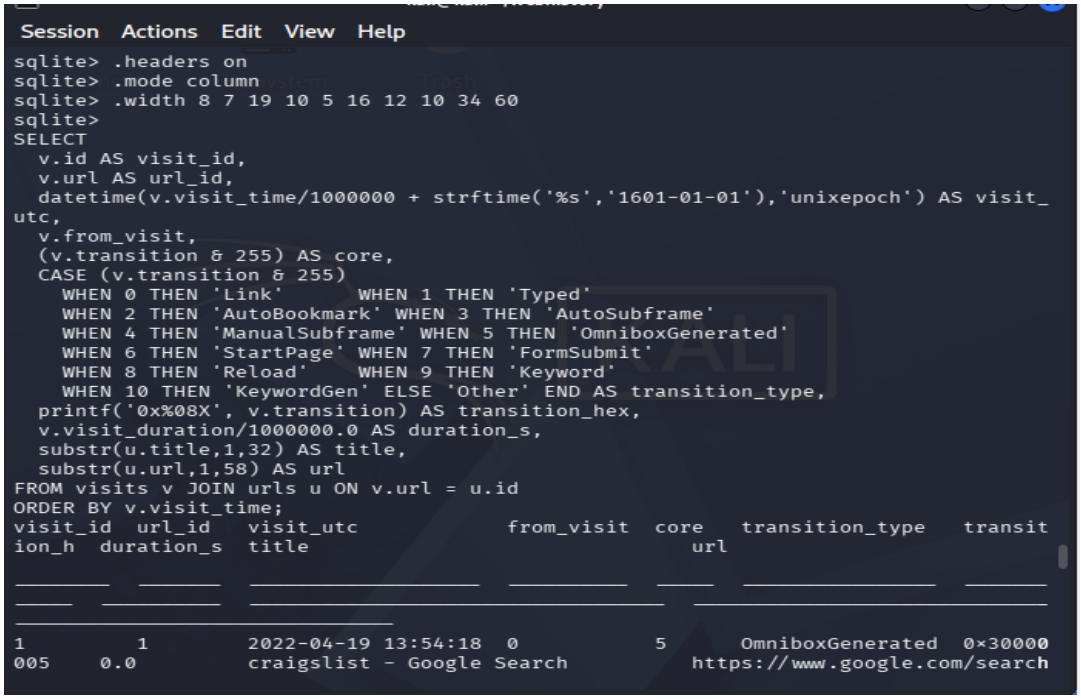
1         1       2022-04-19 13:54:18  0           805306373    0.0          craigsl
ist - Google Search                                 https://www.google.com/search
?q=craigslist&rlz=1C1ONGR_enUS9

                                                    89US989&oq=craigslist&aqs=chr
ome..69i57j46i131i199i433i465i5

                                                    12j0i402l2j0i512l3j0i131i433l
2j0i512.2595j0j15&sourceid=chro

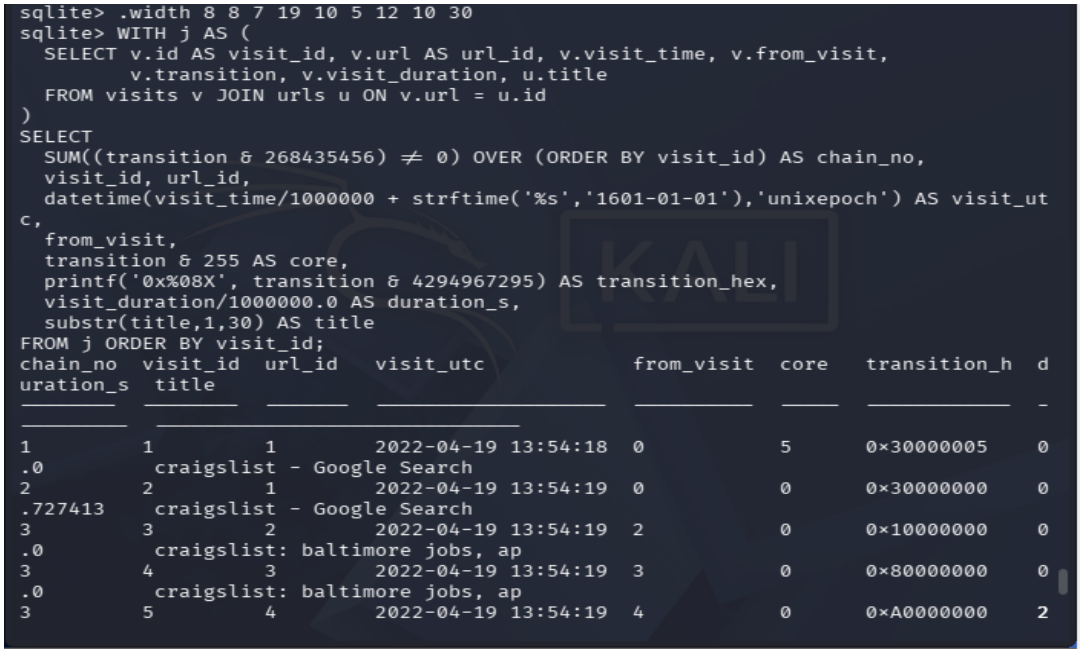
                                                    me&ie=UTF-8
```

Figure 8. First visits and urls join with the WebKit-to-UTC conversion.



```
Session  Actions  Edit  View  Help
sqlite> .headers on
sqlite> .mode column
sqlite> .width 8 7 19 10 5 16 12 10 34 60
sqlite>
SELECT
  v.id AS visit_id,
  v.url AS url_id,
  datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS visit_
utc,
  v.from_visit,
  (v.transition & 255) AS core,
  CASE (v.transition & 255)
    WHEN 0 THEN 'Link'      WHEN 1 THEN 'Typed'
    WHEN 2 THEN 'AutoBookmark' WHEN 3 THEN 'AutoSubframe'
    WHEN 4 THEN 'ManualSubframe' WHEN 5 THEN 'OmniboxGenerated'
    WHEN 6 THEN 'StartPage' WHEN 7 THEN 'FormSubmit'
    WHEN 8 THEN 'Reload'    WHEN 9 THEN 'Keyword'
    WHEN 10 THEN 'KeywordGen' ELSE 'Other' END AS transition_type,
  printf('0x%08X', v.transition) AS transition_hex,
  v.visit_duration/1000000.0 AS duration_s,
  substr(u.title,1,32) AS title,
  substr(u.url,1,58) AS url
FROM visits v JOIN urls u ON v.url = u.id
ORDER BY v.visit_time;
visit_id  url_id   visit_utc           from_visit  core   transition_type   transit
ion_h  duration_s  title                              url
________  _______  ___________________  __________  _____  ________________  _______
_____  __________  _____________________________  ____________________________________
_____________________
1         1        2022-04-19 13:54:18  0           5      OmniboxGenerated  0x30000
005    0.0         craigslist - Google Search         https://www.google.com/search
```

Figure 9. Transition decoding query: base type 5 (OmniboxGenerated) and transition 0x30000005 for visit 1.



```
sqlite> .width 8 8 7 19 10 5 12 10 30
sqlite> WITH j AS (
  SELECT v.id AS visit_id, v.url AS url_id, v.visit_time, v.from_visit,
         v.transition, v.visit_duration, u.title
  FROM visits v JOIN urls u ON v.url = u.id
)
SELECT
  SUM((transition & 268435456) != 0) OVER (ORDER BY visit_id) AS chain_no,
  visit_id, url_id,
  datetime(visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS visit_ut
c,
  from_visit,
  transition & 255 AS core,
  printf('0x%08X', transition & 4294967295) AS transition_hex,
  visit_duration/1000000.0 AS duration_s,
  substr(title,1,30) AS title
FROM j ORDER BY visit_id;
chain_no  visit_id  url_id   visit_utc           from_visit  core   transition_h  d
uration_s  title
________  ________  _______  ___________________  __________  _____  ____________  -
_________  ______________________________
1         1         1        2022-04-19 13:54:18  0           5      0x30000005    0
.0         craigslist - Google Search
2         2         1        2022-04-19 13:54:19  0           0      0x30000000    0
.727413    craigslist - Google Search
3         3         2        2022-04-19 13:54:19  2           0      0x10000000    0
.0         craigslist: baltimore jobs, ap
3         4         3        2022-04-19 13:54:19  3           0      0x80000000    0
.0         craigslist: baltimore jobs, ap
3         5         4        2022-04-19 13:54:19  4           0      0xA0000000    2
```

Figure 10. Chain-numbered timeline query and its first rows (chain 3 covers visits 3-5).

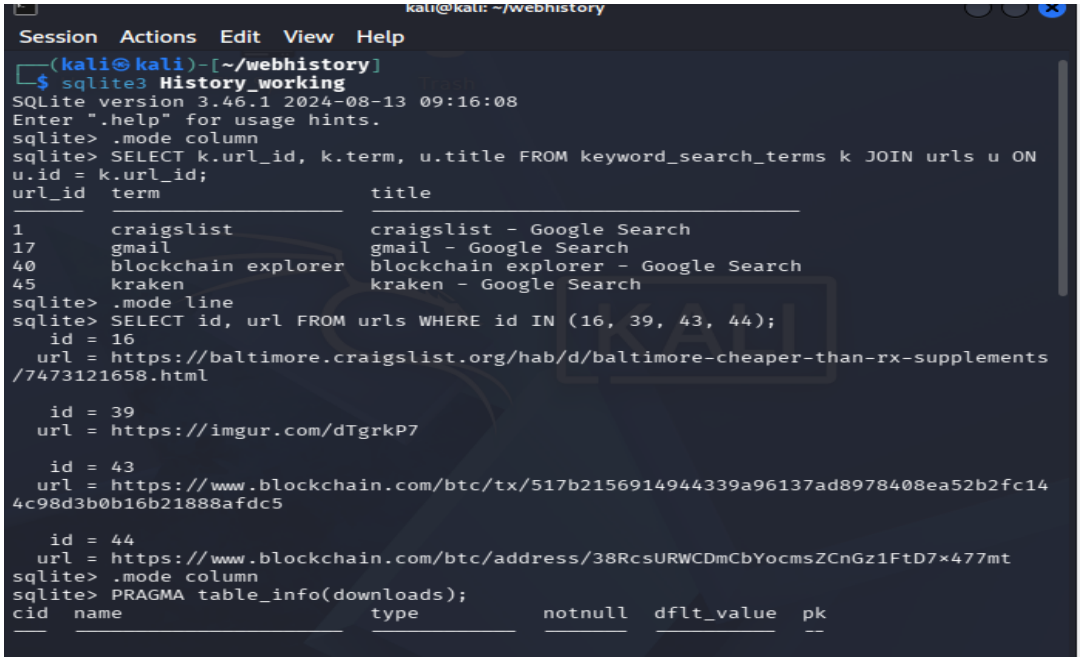


Figure 11. Search terms and the full URLs of urls 16, 39, 43 and 44.

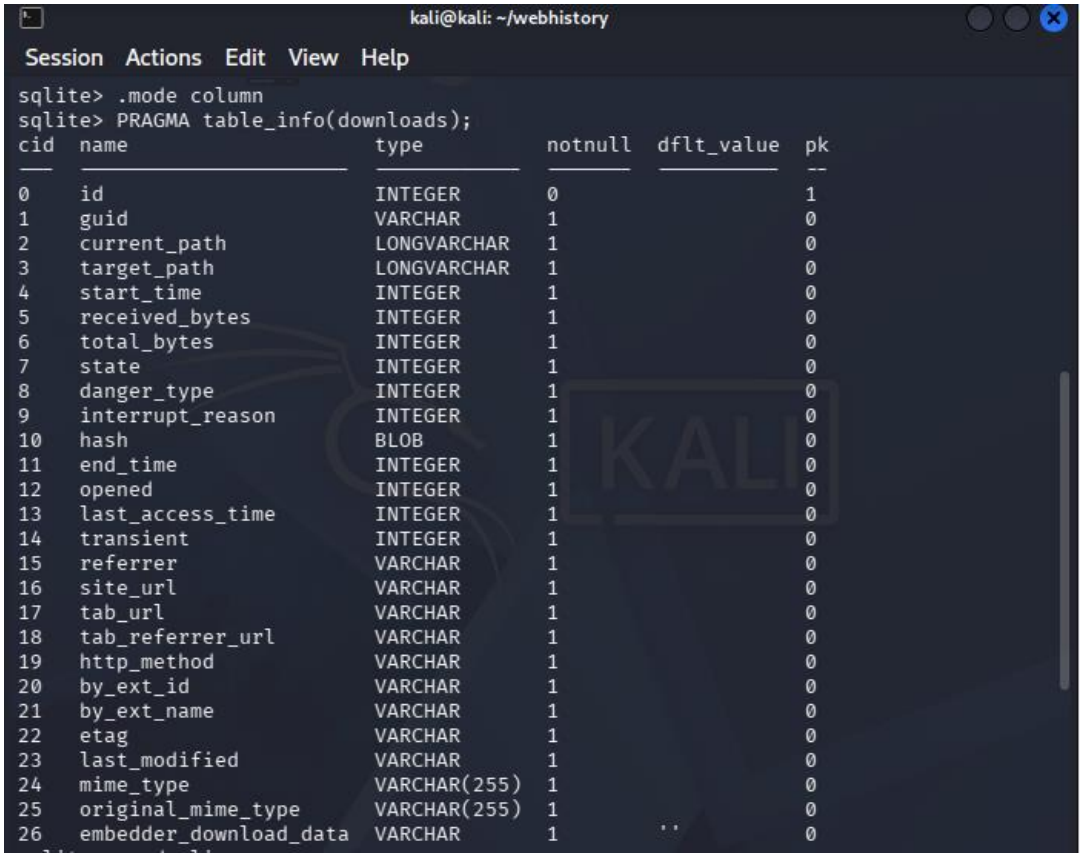


Figure 12. Structure of the downloads table.

```
sqlite> .mode line
sqlite> SELECT id, target_path, total_bytes, state, danger_type, interrupt_reason, d
atetime(start_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS start_utc,
datetime(end_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS end_utc, tab
_url, mime_type, hex(hash) AS sha256_hex FROM downloads;
             id = 1
    target_path = C:\Users\FSCS_User\Desktop\proof_of_payment.png
    total_bytes = 33844
          state = 1
    danger_type = 0
interrupt_reason = 0
       start_utc = 2022-04-19 14:56:58
         end_utc = 2022-04-19 14:57:10
         tab_url = https://imgur.com/dTgrkP7
       mime_type = image/png
      sha256_hex =
sqlite> SELECT * FROM downloads_url_chains;
         id = 1
chain_index = 0
        url = https://i.imgur.com/dTgrkP7.png
sqlite>
```

Figure 13. Download record 1 and its URL chain.

```
Session  Actions  Edit  View  Help
sqlite> .mode line
sqlite> SELECT id, target_path, total_bytes, state, danger_type, interrupt_reason, d
atetime(start_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS start_utc,
datetime(end_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS end_utc, tab
_url, mime_type, hex(hash) AS sha256_hex FROM downloads;
             id = 1
    target_path = C:\Users\FSCS_User\Desktop\proof_of_payment.png
    total_bytes = 33844
          state = 1
    danger_type = 0
interrupt_reason = 0
       start_utc = 2022-04-19 14:56:58
         end_utc = 2022-04-19 14:57:10
         tab_url = https://imgur.com/dTgrkP7
       mime_type = image/png
      sha256_hex =
sqlite> SELECT * FROM downloads_url_chains;
         id = 1
chain_index = 0
        url = https://i.imgur.com/dTgrkP7.png
sqlite> .mode line
sqlite> SELECT id, title FROM urls WHERE id IN (16, 34, 37, 42, 43, 44, 50, 51, 54,
55);
   id = 16
title = cheaper than Rx supplements - health and beauty - by owner - ...

   id = 34
title = cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail

   id = 37
title = cheaper than Rx supplements -- txid: 517b2156914944339a96137ad8978408ea52b2f
c144c98d3b0b16b21888afdc5 - unsub.fscs@gmail.com - Gmail
```

Figure 14. Download record 1 and full titles for urls 16, 34 and 37.

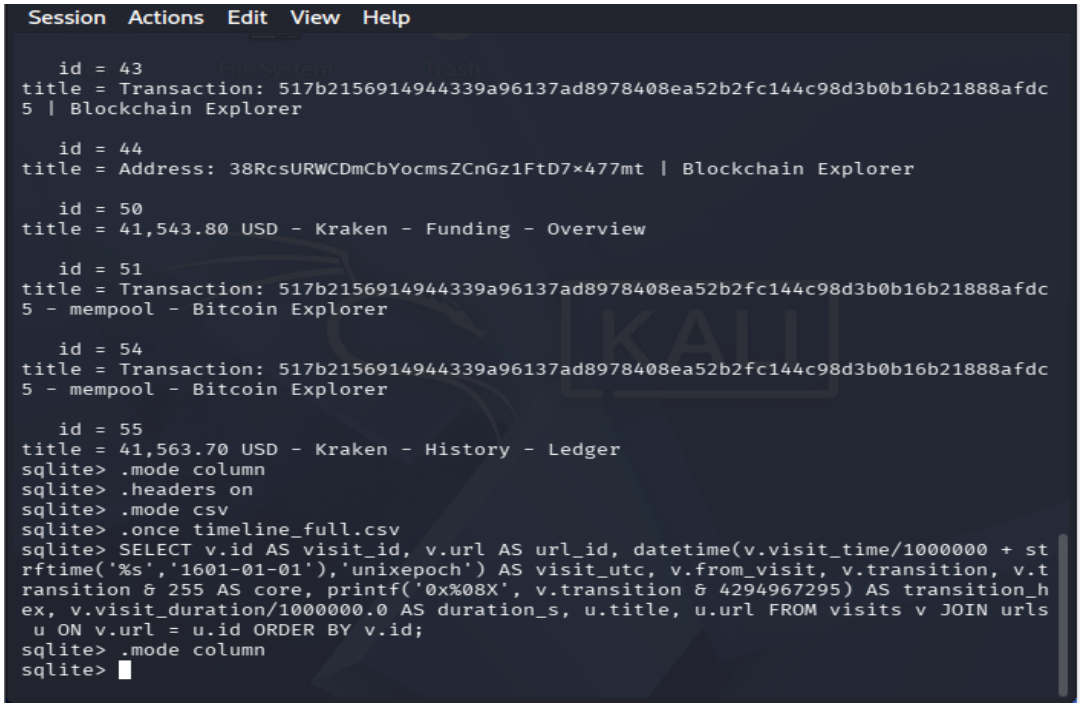


Figure 15. Full titles for urls 42-55 and the timeline_full.csv export command.

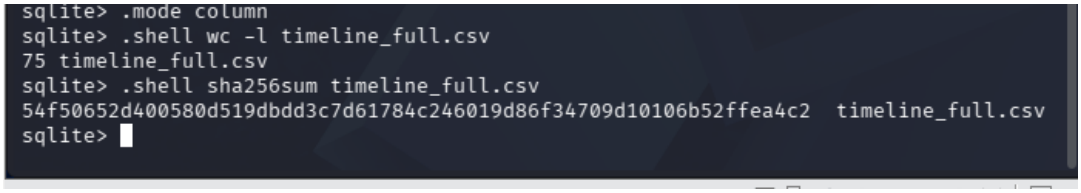


Figure 16. timeline_full.csv line count (75) and SHA-256.

Appendix B: All 74 visits

All times are 2022-04-19 UTC. Full titles and URLs are in `timeline_full.csv`.

Visit	URL	UTC	From	Transition	Dur (s)	Page
1	1	13:54:18	0	0x30000005	0.000	Google search 'craigslist' (omnibox)
2	1	13:54:19	0	0x30000000	0.727	Same results page
3	2	13:54:19	2	0x10000000	0.000	www.craigslist.org (chain start)
4	3	13:54:19	3	0x80000000	0.000	geo.craigslist.org (server redirect)
5	4	13:54:19	4	0xA0000000	2.465	baltimore.craigslist.org (redirect, chain end)
6	5	13:54:22	5	0x10000000	0.000	accounts.craigslist.org/login/home
7	6	13:54:22	6	0xA0000000	17.457	Craigslist log-in page
8	5	13:54:39	7	0x30000007	23.439	Form submit; 'craigslist account'
9	2	13:55:03	8	0x10000000	0.000	www.craigslist.org
10	4	13:55:03	9	0xA0000000	2.664	baltimore.craigslist.org

11	7	13:55:05	10	0x10000000	0.000	post.craigslist.org/c/bal, 'choose type'
12	8	13:55:05	11	0x80000000	0.000	'posting confirmation' URL (mid-redirect)
13	9	13:55:05	12	0xA0000000	4.273	`?s=type`, 'choose type'
14	10	13:55:10	13	0x30000007	5.599	`?s=cat`, 'choose category'
15	11	13:55:15	14	0x30000007	177.675	`?s=edit`, 'create posting'
16	12	13:58:13	15	0x30000007	4.390	`?s=geoverify`, 'add map'
17	13	13:58:17	16	0x30000007	1.649	`?s=editimage`, 'create posting'
18	14	13:58:19	17	0x30000007	3.200	`?s=preview`, 'create posting'
19	8	13:58:22	18	0x30000007	4.989	'posting confirmation'
20	15	13:58:27	19	0x30000000	5.597	`manage/7473121658`
21	16	13:58:31	0	0x30000000	0.000	Ad page 'cheaper than Rx supplements'
22	17	13:58:38	0	0x32000005	0.000	Google search 'gmail' (omnibox)
23	17	13:58:39	0	0x30000000	0.843	Same results page
24	18	13:58:40	23	0x10000000	0.000	google.com/gmail/
25	19	13:58:40	24	0x80000000	0.000	mail.google.com/mail/ (title 'Inbox (12)')
26	20	13:58:40	25	0x80000000	0.000	mail.google.com/mail/u/0/
27	21	13:58:40	26	0x80000000	0.000	accounts.google.com ServiceLogin
28	22	13:58:40	27	0x80000000	0.000	mail.google.com/intl/en/mail/help/about.html
29	23	13:58:40	28	0x80000000	0.000	www.google.com/intl/en/mail/help/about.html
30	24	13:58:40	29	0xA0000000	0.000	google.com/gmail/about/
31	25	13:58:43	0	0x10000000	0.000	AccountChooser signinchooser
32	26	13:58:43	31	0x80000000	0.000	AccountChooser?service=mail
33	27	13:58:43	32	0xA0000000	0.000	ServiceLogin?continue=
34	27	13:58:44	0	0x30000000	0.000	ServiceLogin?continue= (again)
35	28	13:58:44	0	0x30000000	0.000	signin/v2/identifier
36	29	13:59:26	0	0x30000000	2.968	signin/v2/challenge/pwd
37	30	13:59:29	36	0x10000000	0.000	CheckCookie?continue= (title 'Inbox (12)')
38	31	13:59:29	37	0x80000000	0.000	mail.google.com/accounts/SetOSID
39	32	13:59:29	38	0x80000000	0.000	accounts.youtube.com/accounts/SetSID
40	19	13:59:29	39	0x80000000	0.000	mail.google.com/mail/
41	20	13:59:29	40	0x80000000	0.000	mail.google.com/mail/u/0/
42	33	13:59:31	41	0x60000000	0.000	mail/u/0/#inbox (client redirect, chain end)
43	34	14:04:12	0	0x10000000	0.000	Message 'cheaper than Rx supplements - order'
44	35	14:04:35	43	0x40000000	0.000	`?compose=` URL

45	36	14:04:46	44	0x40000000	0.000	Second `?compose=` URL
46	34	14:04:52	45	0x60000000	0.000	Back to the 'order' message
47	33	14:05:59	0	0x30000000	0.000	Inbox
48	37	14:56:47	0	0x10000000	0.000	Message 'cheaper than Rx supplements -- txid: ...'
49	38	14:56:52	0	0x10000000	0.000	google.com/url?q=https://imgur.com/dTgrkP7...
50	39	14:56:52	49	0xA0000000	18.926	imgur.com/dTgrkP7
51	40	14:58:59	0	0x32000005	0.000	Google search 'blockchain explorer' (omnibox)
52	40	14:59:00	0	0x30000000	1.026	Same results page
53	41	14:59:01	52	0x30000000	0.000	blockchain.com/explorer
54	41	14:59:02	0	0x30000000	18.679	blockchain.com/explorer
55	42	14:59:06	54	0x10000000	0.000	blockchain.com/search?search=517b...
56	43	14:59:06	55	0xA0000000	0.000	`btc/tx/517b...` (redirect, chain end)
57	43	14:59:06	54	0x30000000	0.000	`btc/tx/517b...` (single-visit chain)
58	44	14:59:21	54	0x30000000	0.000	`btc/address/38Rcs...`
59	45	14:59:38	0	0x32000005	0.000	Google search 'kraken' (omnibox)
60	45	14:59:39	0	0x30000000	7.704	Same results page
61	46	14:59:47	60	0x30000000	0.000	kraken.com/en-us/sign-in
62	46	14:59:47	0	0x30000000	0.000	kraken.com/en-us/sign-in
63	47	15:00:21	0	0x30000000	104.425	kraken.com/en-us/device-approval
64	48	15:01:11	63	0x30000000	0.000	`/u/trade?i=true` (empty title)
65	49	15:01:12	63	0x30000000	0.000	`/u/instant`
66	50	15:01:21	63	0x30000000	0.000	`/u/funding` (title 41,543.80 USD)
67	51	15:01:31	0	0x10000000	0.000	kraken.com/redirect?url=...mempool...
68	52	15:01:33	67	0x40000000	0.000	kraken.com/redirect?t=... (http)
69	53	15:01:33	68	0x80000000	0.000	kraken.com/redirect?t=... (https)
70	54	15:01:33	69	0xA0000000	0.000	mempool.space/tx/517b...
71	55	15:02:05	63	0x30000000	0.000	`/u/history/ledger` (title 41,563.70 USD)
72	56	15:02:32	48	0x40000000	0.000	`?compose=` URL
73	57	15:02:44	72	0x40000000	0.000	Second `?compose=` URL
74	37	15:02:56	73	0x60000000	0.000	Back to the 'txid' message

Appendix C: SQL queries

```sql
-- Tables
SELECT name FROM sqlite_master WHERE type = 'table' ORDER BY name;

-- Structures
PRAGMA table_info(urls);
PRAGMA table_info(visits);
PRAGMA table_info(downloads);

-- Addresses
SELECT id, url, title, visit_count FROM urls ORDER BY id;

-- Chain-numbered timeline
WITH j AS (
  SELECT v.id AS visit_id, v.url AS url_id, v.visit_time, v.from_visit,
         v.transition, v.visit_duration, u.title
  FROM visits v JOIN urls u ON v.url = u.id
)
SELECT
  SUM((transition & 268435456) != 0) OVER (ORDER BY visit_id) AS chain_no,
  visit_id, url_id,
  datetime(visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS
visit_utc,
  from_visit,
  transition & 255 AS core,
  printf('0x%08X', transition & 4294967295) AS transition_hex,
  visit_duration/1000000.0 AS duration_s,
  substr(title,1,30) AS title
FROM j ORDER BY visit_id;

-- Search terms
SELECT k.url_id, k.term, u.title FROM keyword_search_terms k JOIN urls u ON
u.id = k.url_id;

-- Full URLs and titles
SELECT id, url FROM urls WHERE id IN (16, 39, 43, 44);
SELECT id, title FROM urls WHERE id IN (16, 34, 37, 42, 43, 44, 50, 51, 54,
55);


,
  v.visit_duration/1000000.0 AS duration_s, u.title, u.url
FROM visits v JOIN urls u ON v.url = u.id ORDER BY v.id;
```

```
-- Downloads
SELECT id, target_path, total_bytes, state, danger_type, interrupt_reason,
  datetime(start_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS
start_utc,
  datetime(end_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS
end_utc,
  tab_url, mime_type, hex(hash) AS sha256_hex
FROM downloads;
SELECT * FROM downloads_url_chains;

-- Full export
.headers on
.mode csv
.once timeline_full.csv
SELECT v.id AS visit_id, v.url AS url_id,
  datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS
visit_utc,
  v.from_visit, v.transition, v.transition & 255 AS core,
  printf('0x%08X', v.transition & 4294967295) AS transition_hex
```

Appendix D: Evidence log

ID	Source record	Finding	Why it matters	Figure
E01	History file; ls, file, sha256sum	196,608 bytes; SHA-256 b991b0fa...3fd19; original read-only; working copy has the same hash	Sets the integrity baseline for every later finding	1-3
E02	sqlite_master; PRAGMA table_info on urls, visits, downloads	17 tables; five relevant: urls, visits, downloads, downloads_url_chains, keyword_search_terms	Scopes the examination and shows the field names I relied on	4, 5, 7, 12
E03	visits joined to urls	74 visits from 13:54:18 to 15:02:56 UTC; base types 0, 5 and 7 only; chains decoded from the qualifier bits	Fixes the time basis and the navigation method	8-10; App. B
E04	keyword_search_terms, url IDs 1, 17, 40, 45	Terms: craigslist, gmail, blockchain explorer, kraken	Corroborates the omnibox visits 1, 22, 51 and 59	11
E05	Visits 11-19; urls 7-14 and 8	Posting steps from 'choose type' to 'posting confirmation'; visit 15 lasted 177.67 s	Shows a posting workflow in order	6; App. B
E06	Visits 20-21; urls 15-16	manage/7473121658 and an ad URL ending in 7473121658.html	Links the management page to the ad by post number	6, 11
E07	Visits 22-42; urls 17-33	Gmail search, sign-in pages and inbox titles naming	Shows access to the mailbox	App. B

		unsub.fscs@gmail.com		
E08	Visits 43-47; urls 33-36	Message 'cheaper than Rx supplements - order'; two `?compose=` URLs	Shows messages opened and compose windows, not sent mail	14; App. B
E09	Visits 48-50; urls 37-39; downloads 1; downloads_url_chains 1	txid message; imgur page; `proof_of_payment.png`, 33,844 bytes, from `i.imgur.com/dTgrkP7.png`; empty hash	Shows a saved image and the route to it	11, 13, 14
E10	Visits 51-58; urls 40-44	Explorer search, TXID lookup and address page	Shows the TXID checked on a public explorer	11; App. B
E11	Visits 59-71; urls 45-55	Kraken sign-in, device approval, funding, redirect to mempool.space, ledger	Shows a Kraken session and the same TXID opened through it	15; App. B
E12	Visits 72-74; urls 56, 57, 37	Two more compose URLs and a return to the txid message	Shows the last recorded activity	App. B
E13	Urls 37, 42, 43, 51, 54	The same 64-character TXID in five titles across Gmail, Blockchain.com and mempool.space	Ties the email, the explorer lookup and the Kraken redirect together	14, 15
E14	`timeline_full.csv`	75 lines; SHA-256 `54f50652...ffea4c2`; generated from `History_working`	Makes my timeline reproducible and verifiable	15, 16